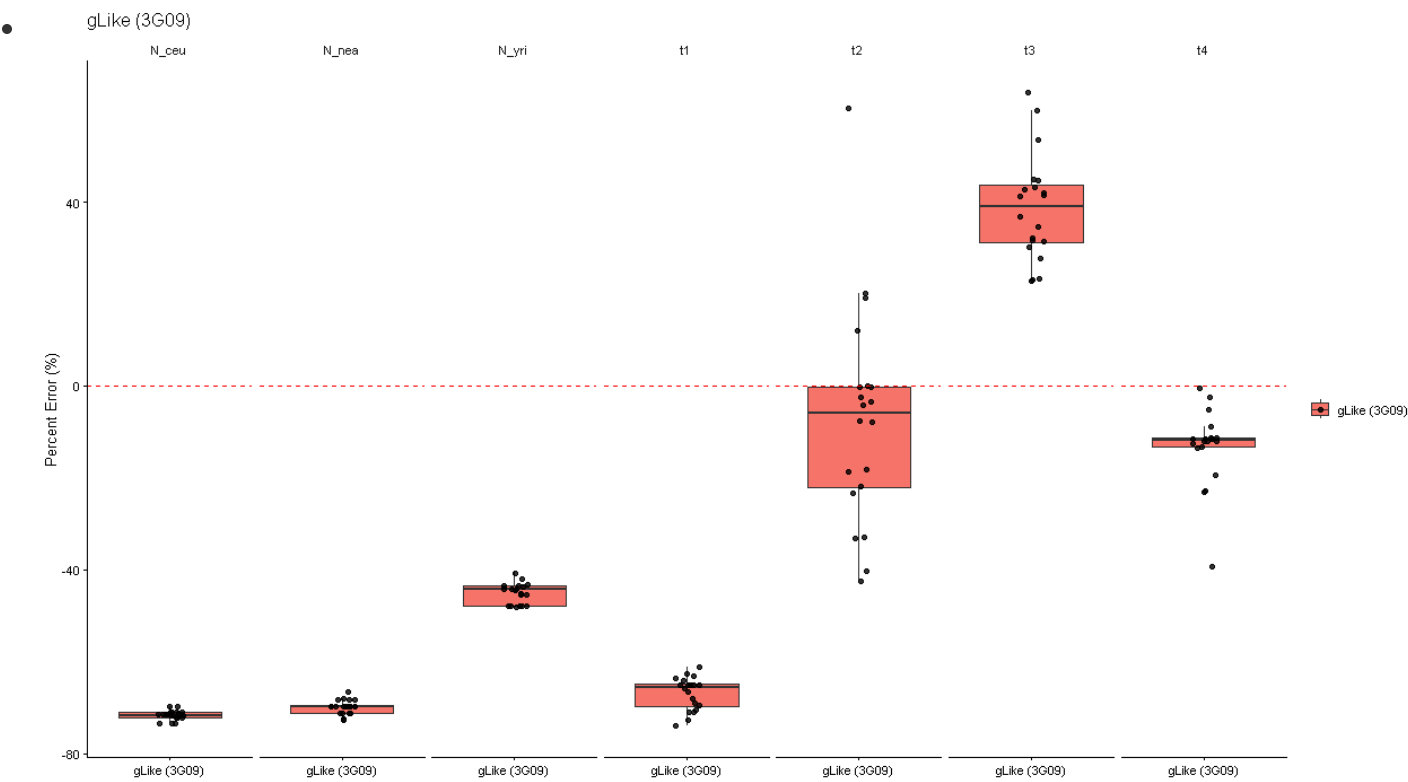


Summary

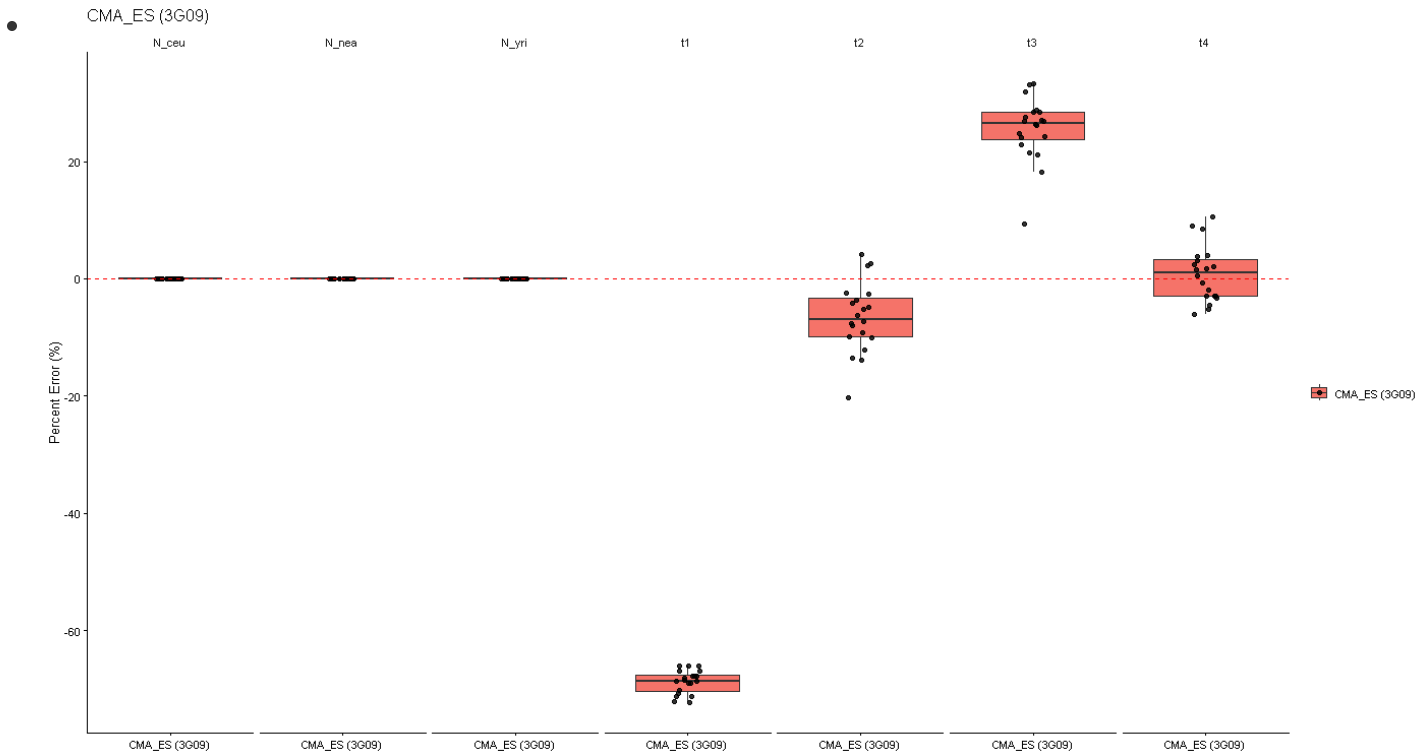
- Re-ran estimates for NH and 3G09 (without migration) after fiddling with code
- Percentage error plots horizontally arranged on the same y-axis

Results

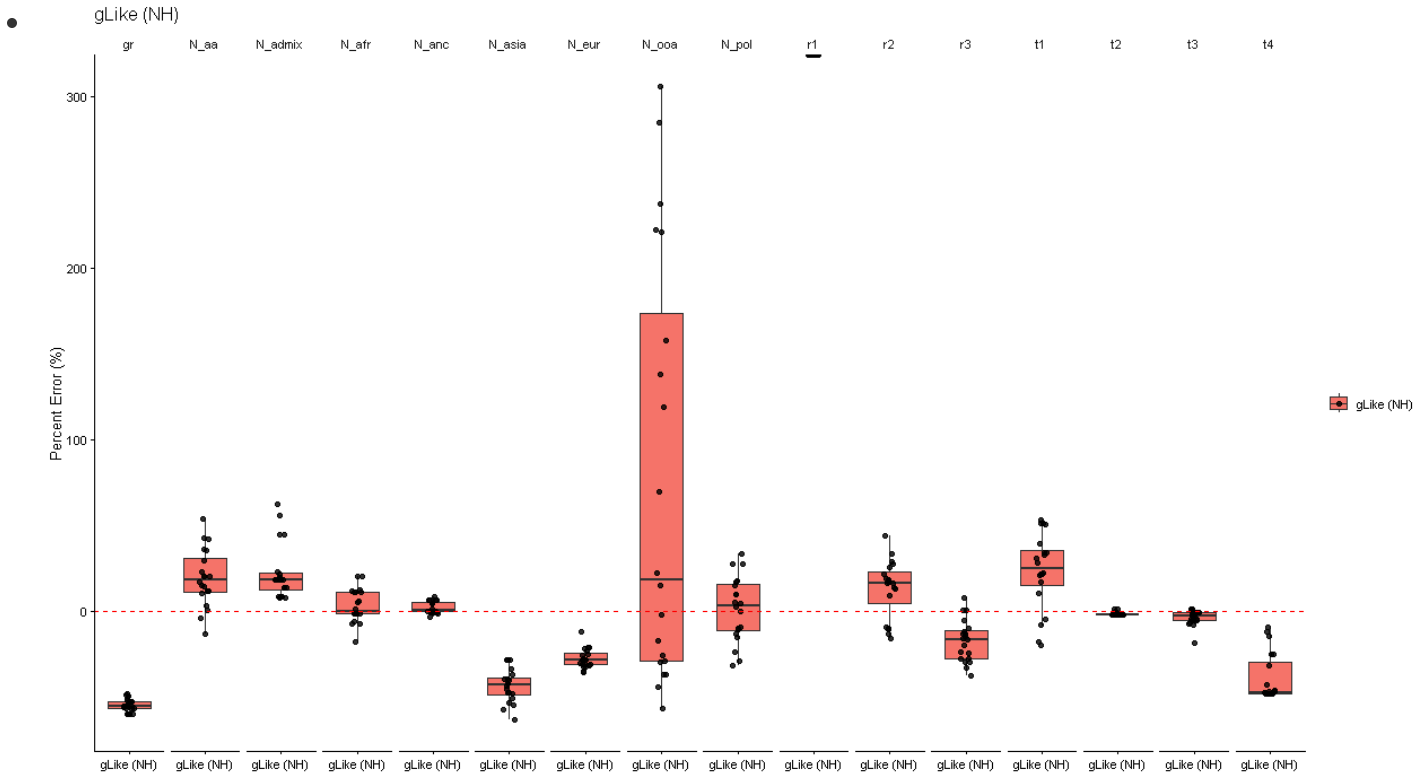
3G09 (gLike)



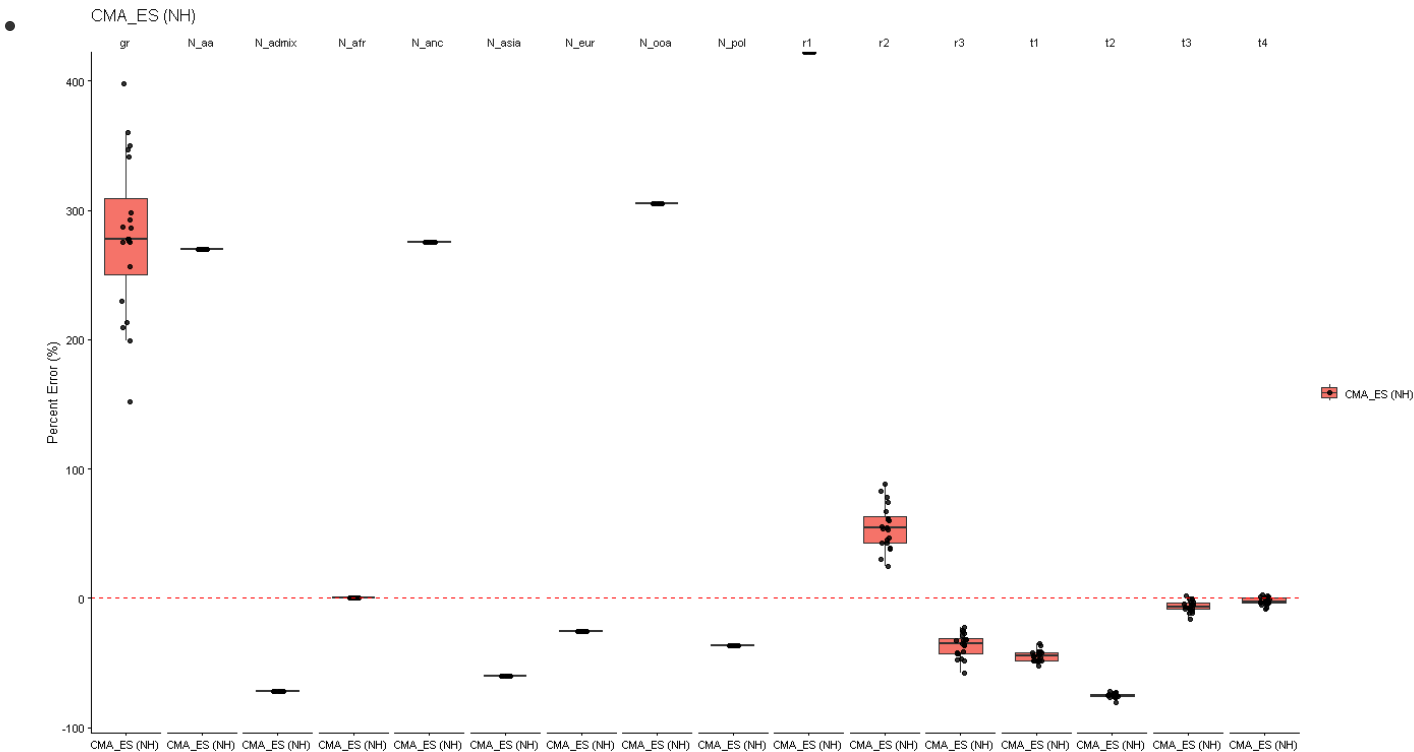
3G09 (CMA-ES)



NH (gLike)



NH (CMA-ES)



Thoughts

- Something that stands out to me is the relative accuracy of inferring ancestral population size for CMA-ES in the 3G09 scenario, but inaccuracy for inferring ancestral population sizes in the NH scenario. This relationship is flipped for gLike.
- Arbitrarily, the initial condition I set for any ancestral population size is N_Anc = 10,000. It turns out that CMA-ES doesn't like to move any of the population size parameters:

```
> CMA_ES_3G09_params
file NUM_TREES N_ceu N_nea N_yri
1 ntree_10_rep_10_simulate_ARG.log 10 10000.086 10000.041 10000.263
2 ntree_10_rep_11_simulate_ARG.log 10 10000.062 9999.853 9999.987
3 ntree_10_rep_12_simulate_ARG.log 10 10000.024 9999.899 10000.076
4 ntree_10_rep_13_simulate_ARG.log 10 10000.121 10000.024 9999.951
5 ntree_10_rep_14_simulate_ARG.log 10 9999.989 9999.966 9999.918
6 ntree_10_rep_15_simulate_ARG.log 10 10000.028 9999.969 9999.964
7 ntree_10_rep_16_simulate_ARG.log 10 10000.019 10000.083 9999.918
8 ntree_10_rep_17_simulate_ARG.log 10 10000.188 10000.030 9999.946
9 ntree_10_rep_18_simulate_ARG.log 10 9999.993 10000.051 9999.883
10 ntree_10_rep_19_simulate_ARG.log 10 9999.985 10000.009 10000.067
11 ntree_10_rep_1_simulate_ARG.log 10 10000.005 9999.941 10000.027
12 ntree_10_rep_20_simulate_ARG.log 10 9999.915 9999.942 10000.102
13 ntree_10_rep_2_simulate_ARG.log 10 9999.912 9999.908 10000.017
14 ntree_10_rep_3_simulate_ARG.log 10 9999.969 9999.914 10000.107
15 ntree_10_rep_4_simulate_ARG.log 10 9999.941 9999.939 9999.873
16 ntree_10_rep_5_simulate_ARG.log 10 10000.002 10000.059 10000.029
17 ntree_10_rep_6_simulate_ARG.log 10 10000.124 9999.999 10000.057
18 ntree_10_rep_7_simulate_ARG.log 10 10000.004 9999.936 10000.042
19 ntree_10_rep_8_simulate_ARG.log 10 10000.002 9999.933 9999.854
20 ntree_10_rep_9_simulate_ARG.log 10 9999.926 10000.002 10000.044
```

```

O > CMA_ES_NH_params
file NUM_TREES N_aa N_admix N_afr N_anc N_asia N_eur N_ooa N_pol
1 ntree_10_rep_10_simulate_ARG.log 10 10000.060 10000.060 10000.028 10000.061 10000.071 9999.919 10000.066 10000.149
2 ntree_10_rep_11_simulate_ARG.log 10 9999.943 10000.031 9999.951 10000.048 9999.965 9999.950 10000.041 10000.003
3 ntree_10_rep_12_simulate_ARG.log 10 9999.970 9999.944 10000.193 9999.901 9999.904 10000.086 9999.985 9999.837
4 ntree_10_rep_13_simulate_ARG.log 10 9999.936 9999.948 9999.994 9999.992 10000.027 10000.025 9999.928 10000.067
5 ntree_10_rep_14_simulate_ARG.log 10 9999.849 10000.030 9999.915 10000.078 9999.984 10000.017 9999.976 9999.944
6 ntree_10_rep_15_simulate_ARG.log 10 10000.024 10000.032 9999.956 10000.017 9999.881 9999.998 9999.889 10000.048
7 ntree_10_rep_16_simulate_ARG.log 10 9999.977 10000.029 10000.076 10000.045 10000.076 10000.036 10000.016 9999.865
8 ntree_10_rep_17_simulate_ARG.log 10 10000.054 10000.040 9999.982 9999.966 10000.034 9999.926 10000.093 10000.006
9 ntree_10_rep_18_simulate_ARG.log 10 10000.026 9999.983 9999.971 9999.967 9999.939 10000.021 10000.220 10000.003
10 ntree_10_rep_19_simulate_ARG.log 10 9999.916 10000.165 9999.919 10000.016 10000.125 9999.987 10000.053 10000.071
11 ntree_10_rep_1_simulate_ARG.log 10 9999.929 10000.001 9999.925 9999.974 10000.058 9999.962 10000.041 10000.100
12 ntree_10_rep_20_simulate_ARG.log 10 9999.997 9999.944 9999.947 9999.961 10000.032 9999.975 9999.847 9999.998
13 ntree_10_rep_2_simulate_ARG.log 10 10000.069 10000.096 9999.895 9999.897 9999.961 10000.089 9999.984 10000.006
14 ntree_10_rep_3_simulate_ARG.log 10 10000.003 9999.951 9999.977 9999.997 10000.025 9999.985 10000.084 10000.017
15 ntree_10_rep_4_simulate_ARG.log 10 10000.100 10000.022 10000.015 10000.029 9999.920 9999.904 10000.036 9999.919
16 ntree_10_rep_5_simulate_ARG.log 10 9999.910 10000.016 9999.981 10000.031 10000.039 9999.919 9999.963 9999.995
17 ntree_10_rep_6_simulate_ARG.log 10 9999.858 9999.990 9999.974 9999.878 9999.840 10000.061 10000.068 10000.086
18 ntree_10_rep_7_simulate_ARG.log 10 10000.019 9999.949 9999.864 10000.005 10000.004 9999.950 10000.042 10000.044
19 ntree_10_rep_8_simulate_ARG.log 10 10000.117 9999.928 9999.936 10000.008 9999.972 10000.009 9999.998 10000.012
20 ntree_10_rep_9_simulate_ARG.log 10 9999.943 9999.924 10000.014 10000.003 10000.033 9999.946 9999.834 10000.015

```

- Across all 20 replicates for both demographic scenarios there is essentially no movement for inference of N_Anc for any of the subpopulations.